

Notes on Geometric Optics

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11th August 2026

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1 Fermat's Principle

1.1 Optical Path Length (OPL)

Measures how long a path **feels** to a light wave. In a medium, a light wave moves slower compared to vacuum. Thus OPL for a distance d , in a medium is more than that for vacuum.

Since the speed of light in a medium is n times lower than that in vacuum (where n is the refractive index), light takes the same time to travel ' d ' thickness in the medium as nd in vacuum. Hence, OPL in a medium is given by

$$\text{OPL} = n \times d \quad (1)$$

For a varying refractive index, OPL is given by

$$\text{OPL} = \int n(\vec{r}) dl \quad (2)$$

Using this, we can find the phase shift that a light experiences. If a light wave moves by a distance x , The phase difference is given by

$$\Delta\phi = 2\pi \left(\frac{x}{\lambda} \right) \quad (3)$$

where λ is the wavelength in that medium.

Or in terms of Time Period, we say, if a light wave moves for a time t , The phase difference is given by

$$\Delta\phi = 2\pi \left(\frac{t}{T} \right) \quad (4)$$

where T is the Time Period

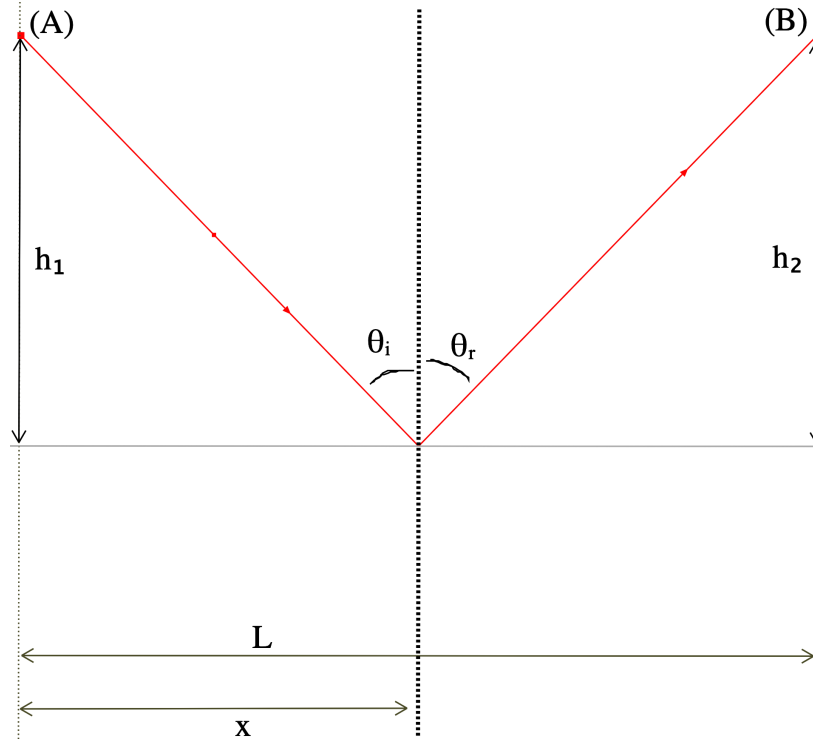
1.2 Definition of Fermat's Principle

Definition 1: According to Fermat's principle, the path that light takes between two points is the one that can be traversed in the least amount of time or has the least optical path length.

This definition is slightly inaccurate because it doesn't take into account the fact that saddle points could exist or that a global maxima or minima may not exist. If this is the case, our definition needs to be modified slightly to...

Definition 2: The path taken by a ray of light between two points is one for which the optical path length (or travel time) is stationary with respect to small variations in the path.

1.3 Law of Reflection using Fermat's Principle



We know that, the optical path length here is the sum of the distances that the light travels to the mirror and back

$$\Rightarrow \text{OPL} = \sqrt{x^2 + h_1^2} + \sqrt{(L-x)^2 + h_2^2}$$

$$\frac{d}{dx}(\text{OPL}) = \frac{x}{\sqrt{x^2 + h_1^2}} - \frac{L-x}{\sqrt{(L-x)^2 + h_2^2}}$$

At stationary point (aka critical point), this first derivative would be zero. That is...

$$\frac{x}{\sqrt{x^2 + h_1^2}} = \frac{L-x}{\sqrt{(L-x)^2 + h_2^2}}$$

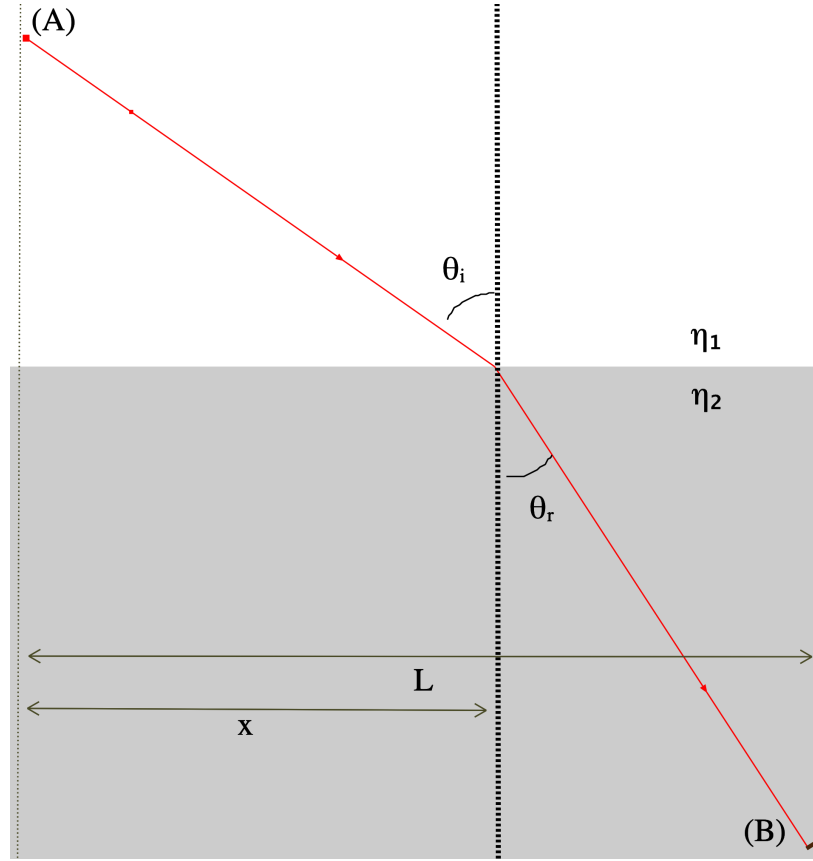
Now, $\frac{x}{\sqrt{x^2 + h_1^2}}$ is nothing but $\sin(\theta_i)$ and $\frac{L-x}{\sqrt{(L-x)^2 + h_2^2}}$ is nothing but $\sin(\theta_r)$. Therefore

$$\sin(\theta_i) = \sin(\theta_r)$$

Since both angles are acute, the only way this is possible is for $\theta_i = \theta_r$.

$$\boxed{\theta_i = \theta_r}$$

1.4 Law of Refraction using Fermat's Principle



Just as before we add up the individual optical path lengths to get the total optical path length. But unlike last time, now we have refractive indices involved and we must use (1)

$$\begin{aligned} \Rightarrow \text{OPL} &= n_1 \sqrt{x^2 + h_1^2} + n_2 \sqrt{(L-x)^2 + h_2^2} \\ \Rightarrow \frac{d}{dx}(\text{OPL}) &= \frac{n_1 x}{\sqrt{x^2 + h_1^2}} - \frac{n_2 (L-x)}{\sqrt{(L-x)^2 + h_2^2}} = 0 \\ \Rightarrow \frac{n_1 x}{\sqrt{x^2 + h_1^2}} &= \frac{n_2 (L-x)}{\sqrt{(L-x)^2 + h_2^2}} \end{aligned}$$

Substituting $\sin(\theta_i)$ for $\frac{x}{\sqrt{x^2 + h_1^2}}$ and $\sin(\theta_r)$ for $\frac{L-x}{\sqrt{(L-x)^2 + h_2^2}}$, we get...

$$\boxed{n_1 \sin(\theta_i) = n_2 \sin(\theta_r)}$$

Hence we derived Snell's law from Fermat's principle.

2 Spherical Refracting Surfaces

2.1 Sign Convention

I use the same sign convention as Ajoy Ghatak. That is...

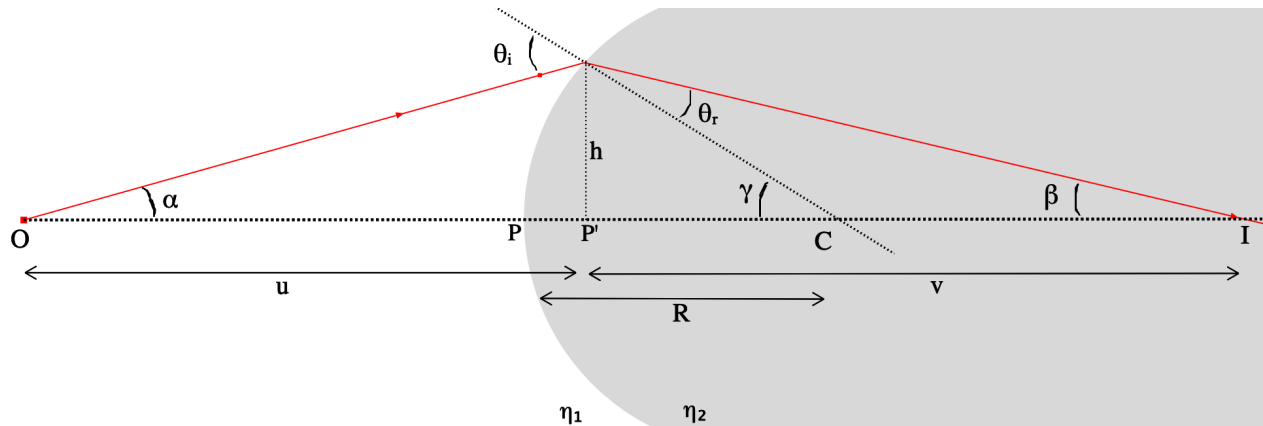
1. All distances on the axis are measured from the middle of the refracting surface. If measured rightwards, that distance is positive and if measured leftwards, that distance is negative.
2. All heights are measured from the axis and it is positive if measured upwards and negative if measured downwards.
3. To give signs to focal length or radius of curvature, we look at on which surface the light is incident first.

Object on the left	–
Object on the right	+
Image on the left	–
Image on the right	+
Anything above Principal axis	+
Anything below Principal axis	–

Table 1: Sign Convention w.r.t to position of lens

2.2 Refraction at a spherical interface

An object is placed in medium-1 at position- O . The light from the object gets incident on the curved surface of the interface of medium-2 and bends towards the normal (assuming $n_2 > n_1$). The point where this refracted ray hits the principal axis is where the image is formed. Object distance is u (sign is negative in this case) and image distance is v (sign is positive). The radius of curvature of the interface is R (sign is positive).



We use an approximation here that we call *paraxial approximation*. Its where we only account

for the rays that hit very close the pole of the lens. When this happens, P and P' coincides. Also, all angles denoted in this diagram then follow small angle approximations.

Firstly, according to Snell's law, $n_1 \sin(\theta_i) = n_2 \sin(\theta_r)$. But since $\sin \theta \approx \theta$,

$$n_1 \theta_i = n_2 \theta_r \quad (5)$$

By exterior angle property,

$$\begin{aligned} \theta_i &= \gamma + \alpha \\ \gamma &= \beta + \theta_r \end{aligned}$$

Taking the ratio of these equations, we get

$$\Rightarrow \frac{\theta_i}{\theta_r} = \frac{\gamma + \alpha}{\gamma - \beta} = \frac{n_2}{n_1} \quad (6)$$

Now, in order to write α , β and γ in terms of distances, I use small angle approximations of trigonometry —

$$\alpha \approx \tan(\alpha) = \frac{h}{-u} \quad (7)$$

$$\beta \approx \tan(\beta) = \frac{h}{v} \quad (8)$$

$$\gamma \approx \tan(\gamma) = \frac{h}{R} \quad (9)$$

For equation (7), the reader who will apply my sign conventions while doing problems will put a negative value for u . The minus sign in front of that u will cancel it out leaving α to be positive just as we wanted.

Substituting these in earlier equation (6) we get,

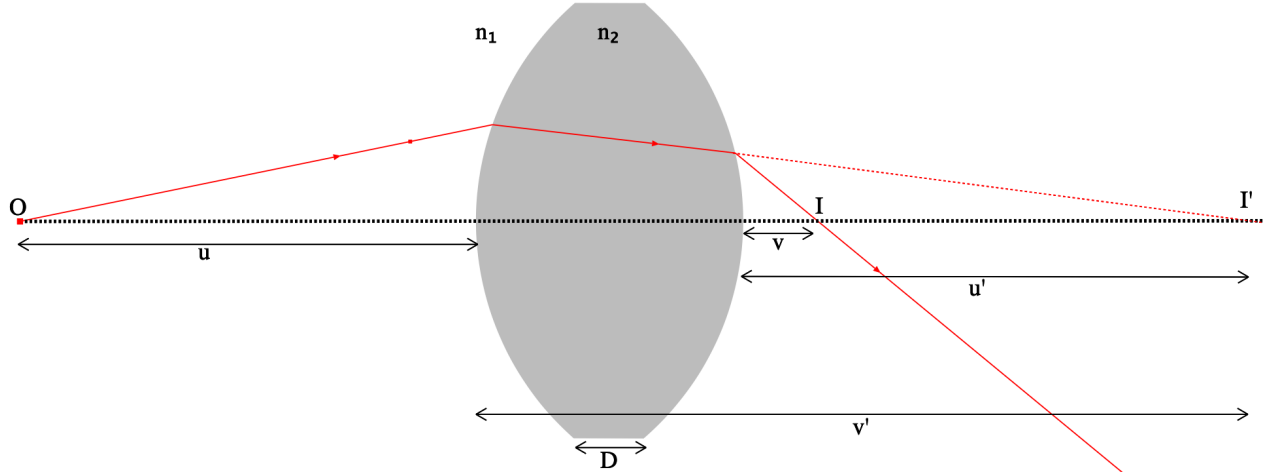
$$\begin{aligned} \frac{\frac{h}{R} - \frac{h}{u}}{\frac{h}{R} - \frac{h}{v}} &= \frac{n_2}{n_1} \\ \Rightarrow n_1 \left(\frac{1}{R} - \frac{1}{u} \right) &= n_2 \left(\frac{1}{R} - \frac{1}{v} \right) \end{aligned}$$

Upon simplification, I arrive at

$$\boxed{\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}} \quad (10)$$

2.3 Thin Lens formula and Image formation

A thin lens can be considered as the composition of two spherical interfaces — One that goes from medium-1 to medium-2 and the other that goes from medium-2 to medium-1. But we also have to consider the thickness of the lens - D .



We can think of it like this – The image that the first interface produces acts as a virtual object for the second interface.

Using equation (10) for the first interface,

$$\frac{n_2}{v'} - \frac{n_1}{u} = \frac{n_2 - n_1}{R_1} \quad (11)$$

For second interface, it would be

$$\begin{aligned} \frac{n_1}{v} - \frac{n_2}{u'} &= \frac{n_1 - n_2}{R_2} \\ \Rightarrow \frac{n_1}{v} - \frac{n_2}{v' - D} &= \frac{n_1 - n_2}{R_2} \end{aligned} \quad (12)$$

Combining equation (11) and (12), we get

$$\frac{1}{v} - \frac{1}{u} = \left(\frac{n_2}{n_1} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) + \frac{n_2 D}{n_1 v' (v' - D)} \quad (13)$$

For thin lens, $D \rightarrow 0$, thus the above equation simplifies to

$$\boxed{\frac{1}{v} - \frac{1}{u} = \left(\frac{n_2}{n_1} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)} \quad (14)$$

This is the equation of a thin lens.

By definition. The focal point is the point of convergence for rays coming from infinity. If we put $u = -\infty$ and $v = f$, we get

$$\boxed{\frac{1}{f} = \left(\frac{n_2}{n_1} - 1\right) \left(\frac{1}{R_1} - \frac{1}{R_2}\right)} \quad (15)$$

This is called Lens maker's formula. Note that the sign of the focal length of a lens not only depends on whether it is convex or concave but also in what medium that lens is placed in. If $n_1 > n_2$, a convex lens acts as a diverging lens and concave lens acts as a converging lens.

From the above expressions (14) and (15), I also realize the lens formula...

$$\boxed{\frac{1}{f} = \frac{1}{v} - \frac{1}{u}} \quad (16)$$

2.4 Magnification of a Lens

The size of the image formed by a lens could be enlarged or diminished depending on the relative positions of object and image. By convention, a negative magnification denotes a real image and a positive magnification denotes a virtual image. By definition, magnification is the ratio of image height to object height. But by property of similar triangles, we can see that

$$m = \frac{h_i}{h_o} = \frac{v}{u} \quad (17)$$

The above magnification is for transverse or linear magnification. Which compares the height of the object and image perpendicular to the principal axis. But there could also be magnification along the principal axis. This is called longitudinal magnification.

For this, we don't assume putting a rod with some non-zero width on the principal axis but we take a point object, place it on the principal axis at some object distance u (so as to get an image at image distance v) and then move it by du so as to move the image by dv . If we assume the initial and final positions of the object and image as the thickness of the object and image respectively, then $dv/du =$ longitudinal magnification.

Differentiating on both sides of the lens formula (16) with respect to u ,

$$\begin{aligned} -\frac{1}{v^2} \frac{dv}{du} + \frac{1}{u^2} &= 0 \\ \implies -\frac{1}{v^2} \frac{dv}{du} &= -\frac{1}{u^2} \\ \implies \frac{dv}{du} &= \left(\frac{v}{u}\right)^2 \end{aligned}$$

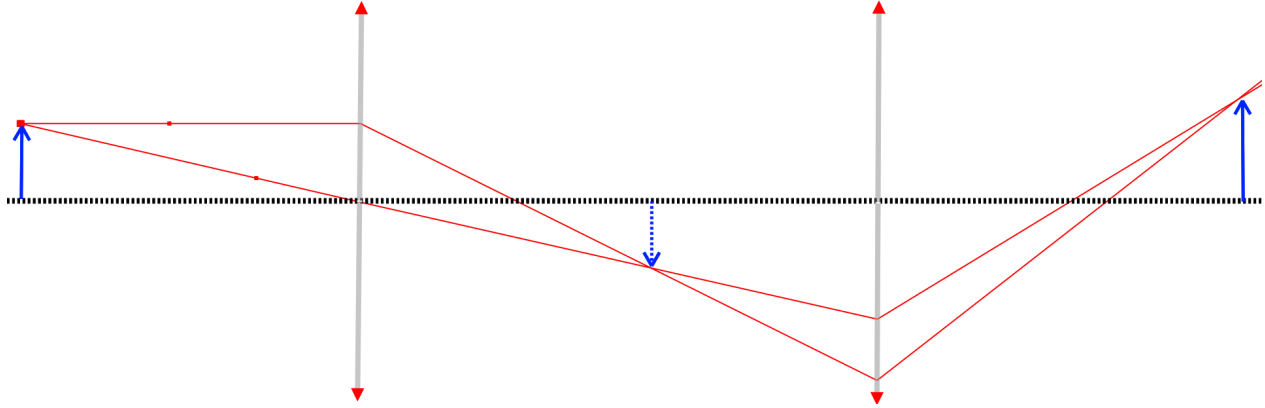
Hence we arrive at the equation for longitudinal magnification. We see why the other magnification is called linear magnification and this is not

$$m_{\text{longitudinal}} = \left(\frac{v}{u}\right)^2 = (m_T)^2 \quad (18)$$

3 Combination of Lenses

3.1 Thin lens combination

We use the image formed by the first lens as the virtual object for the second lens.



First, we use the lens formula (16) to find the position of the first image formation. By subtracting that from the distance between the lenses, I get the magnitude of the object distance for the second lens. Then I apply lens formula again to obtain the final image distance of the object.

If the height of object is h_o , first image is h' and final image is h_i , then

$$m_{\text{net}} = \frac{h_i}{h_o} = \frac{h_i}{h'} \times \frac{h'}{h_o} = m_1 \times m_2 \quad (19)$$

The focal length of the combination is given by

$$\frac{1}{f_{\text{net}}} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2} \quad (20)$$

4 Optical Instruments

4.1 The Human Eye

The human eye contains two major optical components – The pupil which acts as a variable aperture that controls how much light can enter to adjust to different brightness conditions and the cornea which is a lens that focuses the incoming light on the retina which acts as a screen.

To be more specific, the cornea is actually a lens of fixed focal length that takes care of 70% of the focusing. Only the remaining 30% is taken care of by a fluid-like crystalline lens that can have variable thickness and thus variable focal length. But since we are the superior physicists, we shall refer to both the fixed and variable lens as the cornea.

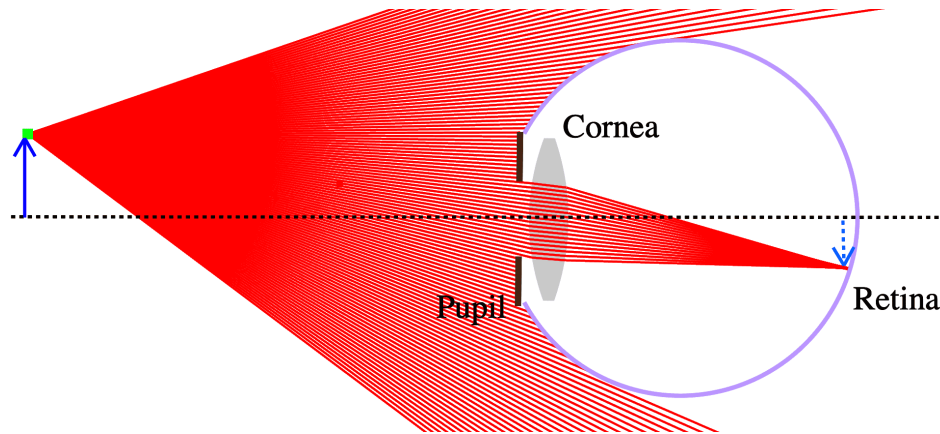


Figure 1: How the Human eye sees

Now, the function of the variable focal length lens is so that all light irrespective of how far the source is, gets focused at the same point. The cornea is most relaxed (highest focal length) when the light is coming from infinity. This happens when we are trying to focus on a distant object. Some people suffer from myopia because their cornea can't relax enough (or their retina is farther) and the light focuses at a point before the retina.

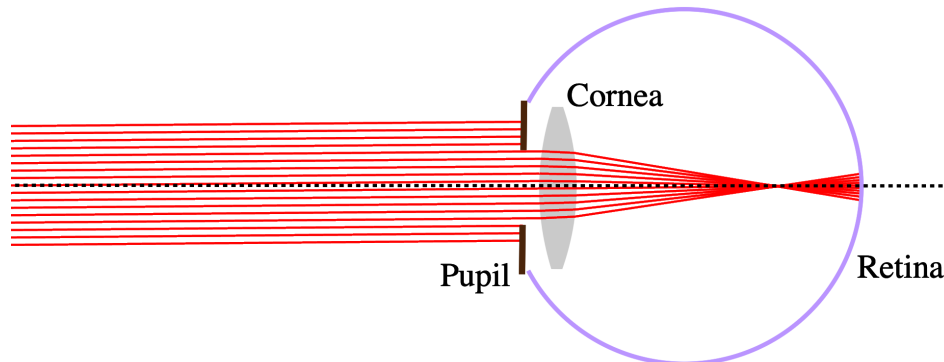


Figure 2: This is how a person with Myopia sees without his glasses

Luckily, we can fix this by placing a concave lens in front of our eyes so that the light slightly diverges before reaching our eyes and the cornea's most relaxed state is now able to focus the light on the retina. In an essence, the concave lens make sure that the light is coming not from far away but from a point nearer.

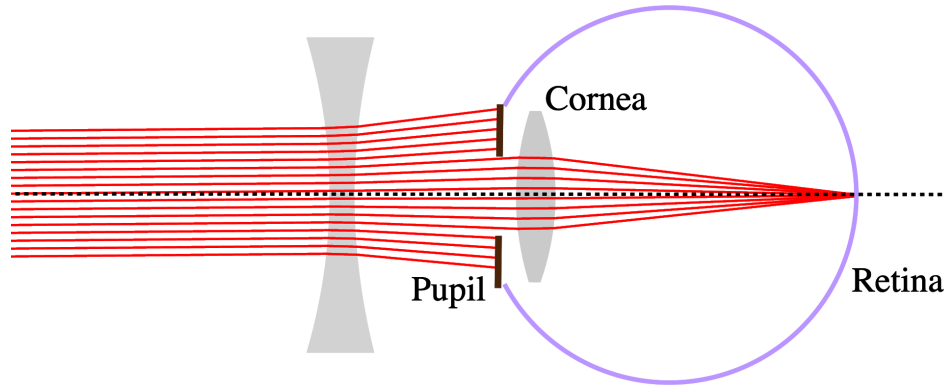
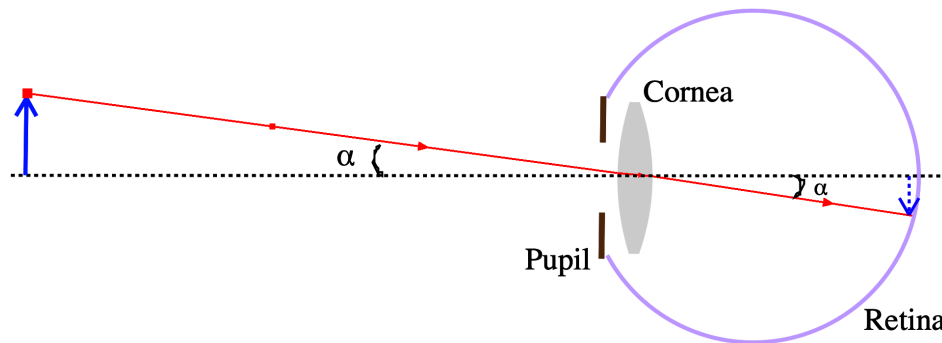


Figure 3: This is how glasses help a person with Myopia see normally

We just saw how the human cornea cannot relax beyond a certain limit. On the other end of the spectrum, the cornea cannot contract beyond a certain point either. To see that, let's bring our object at infinity closer to the eye. To keep the image focused on the retina, the eye-lens must contract more and more (decrease its focal length).



As the object comes closer and closer to the eye, the angle subtended becomes higher and higher. This is what we perceive as the image getting bigger. But we can't bring it closer beyond a certain distance (called the *near-point* of the eye). This is because, although α is still increasing, the light cannot sharply focus onto the retina because we have hit the upper limit of how much our eyes can contract. Without any optical devices, the maximum magnification our eyes occur when object is at near point of the eye (usually 25 cm). That angle subtended then is called α_u where the u stands for 'unaided'.

$$\tan(\alpha_u) = \frac{h_o}{-u} = \frac{h_0}{D}$$

4.2 Simple Microscope

In order to make an object clearly visible past the near point of our eyes, we insert a convex lens between our eye and the object so that it takes care of some of the focusing and our cornea is more relaxed. This is the working principle of a simple microscope.

There are two extreme ways the external lens can be used. One where the eye perceives the object to be infinity (rays reaching the eye are parallel) and the cornea is most relaxed. The second where the eye perceives the object to be at the near point and the eye is most contracted but can still see the image clearly. Any other configuration is between these two extremes.

4.2.1 Virtual object at Infinity

We choose an appropriate convex lens and place it in front of the eye in such a way that the external lens converges the incoming rays into parallel rays such that the eye perceives that as the rays coming from infinity and thus believes that the virtual object is at infinity. The external lens does most of the converging here and the cornea is most relaxed

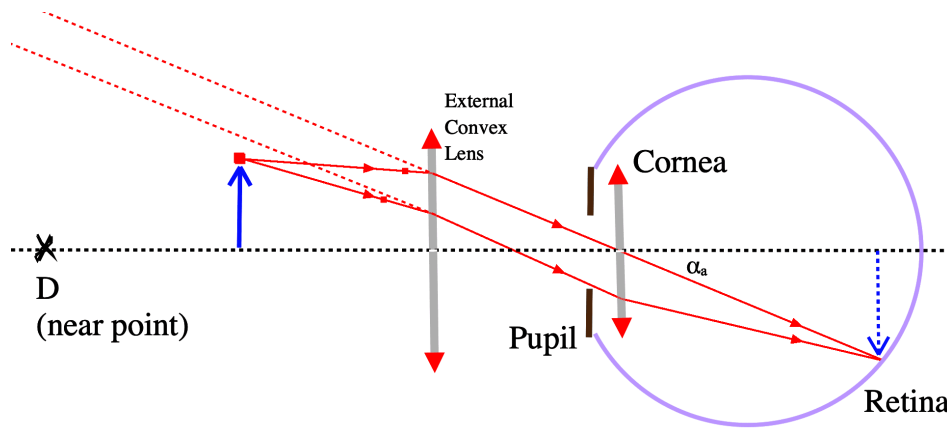


Figure 4: Simple microscope that produces virtual object at infinity for the cornea

The first lens is converging the incoming rays into parallel because the object is at the focus of the external convex lens. Thus the distance from the external lens to object is the focal length of that lens. For practical purposes, we can assume that the virtual object and the actual object form two similar triangles and thus

$$\tan(\alpha_a) = \frac{h'_o \rightarrow \infty}{u' \rightarrow \infty} = \frac{h_o}{f}$$

During the paraxial approximation case, $\tan(\alpha_u) \approx \alpha_u$ and $\tan(\alpha_a) \approx \alpha_a$. And I can write angular magnification as

$$m = \frac{\alpha_a}{\alpha_u} = \frac{h_o/f}{h_o/D} = \frac{D}{f} \quad (21)$$

4.2.2 Virtual object at Near-Point

Similar to the last one, we choose an appropriate convex lens and place it in front of the eye in such a way that the external lens converges the incoming rays into less diverging rays such that the eye perceives that as the rays coming from near point and thus believes that the

virtual object is at the near point. The external lens does part of the converging here but the cornea is most contracted for getting a sharp image.

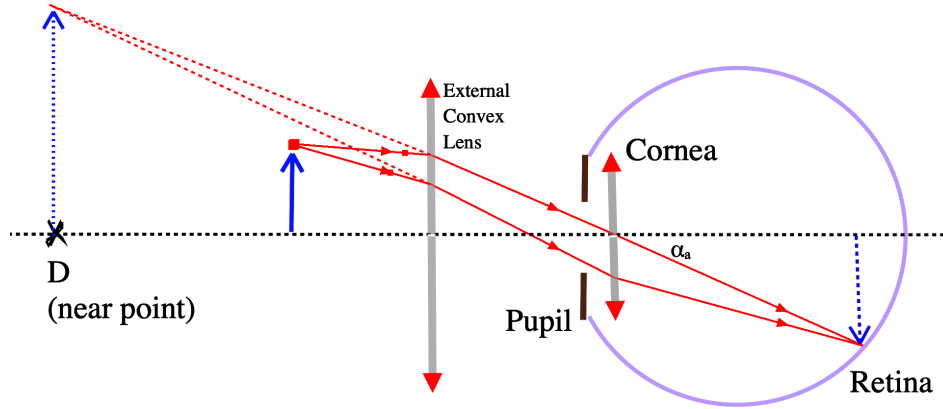


Figure 5: Simple microscope that produces virtual object at the near-point(D) for the cornea

Here, $\alpha_a = \frac{h'_o}{-u'} = \frac{h'_o}{D}$. I can also write $h'_o = \frac{-h_o D}{u}$ and $u = \frac{-Df}{D+f}$. Thus, α_a simplifies to

$$\alpha_a = \frac{h_o(D+f)}{Df}$$

Here too, I can write an expression for magnification

$$m = \frac{\alpha_a}{\alpha_u} = \frac{\frac{h_o(D+f)}{Df}}{\frac{h_o}{D}} = 1 + \frac{D}{f} \quad (22)$$

Thus, given a convex lens and an object, the highest angular magnification is obtained using this near-point configuration.

4.3 Compound Microscope

The magnification produced by a simple microscope can be improved by using a combination of lenses. This is called a compound microscope. It contains two lenses – Objective and eyepiece, separated by a distance 'L'. Once again, there are two orientations to set my compound microscope in – one where the final virtual object forms at infinity (eye relaxed) and the other where it forms at D (eye contracted).

I can write the net magnification as the product of the individual magnifications and use the two cases for eye piece from above (21) and (22). Note that the final **angular** magnification is not the product of the angular magnifications of the individual lenses. This is because we can't define an angular magnification well for the image formed by the objective. Thus, generally for a compound microscope of n -lenses we take the product of linear magnification for $n - 1$ lenses and the angular magnification of the last lens.

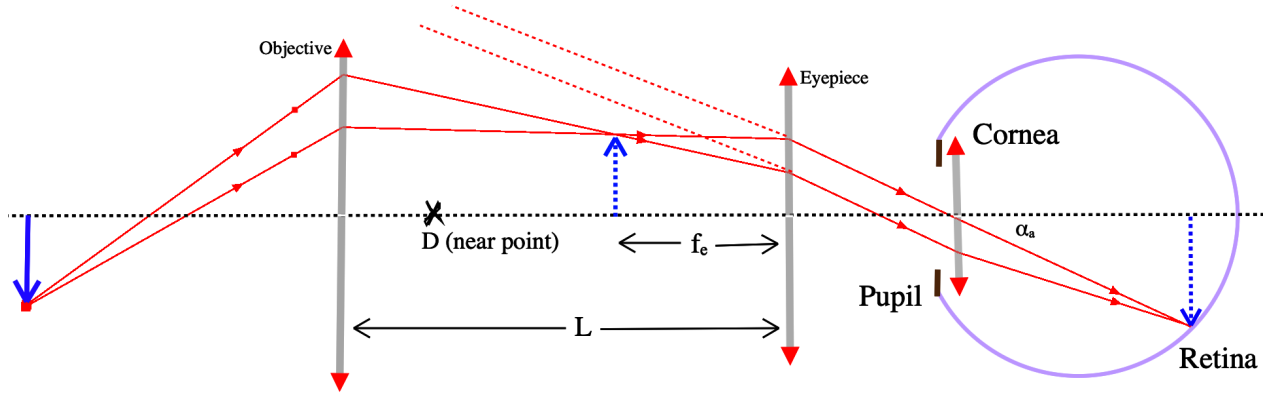


Figure 6: Compound microscope that produces an inverted virtual object at infinity

I can write the lens formula for objective as

$$\frac{1}{f_o} = \frac{1}{v} - \frac{1}{u}$$

multiplying throughout my v , I get

$$\frac{v}{f_o} = 1 - \frac{v}{u}$$

I can write v as $L - f_e$. Since $-v/u$ is nothing but the magnification of the objective,

$$m_o = \frac{L - f_e}{f_o} - 1 = \frac{L - (f_e + f_o)}{f_o}$$

We multiply this with the angular magnification of the eyepiece for the two cases. Thus, by doing $M = m_o \times M_e$, we get...

$$M_\infty = \left(\frac{L - (f_e + f_o)}{f_o} \right) \left(\frac{D}{f_e} \right) \approx \left(\frac{L}{f_o} \right) \left(\frac{D}{f_e} \right)$$

$$M_{\text{near point}} = \left(\frac{L - (f_e + f_o)}{f_o} \right) \left(1 + \frac{D}{f_e} \right) \approx \left(\frac{L}{f_o} \right) \left(1 + \frac{D}{f_e} \right)$$

The approximation is done because usually the distance between the objective and eyepiece is much much greater than the focal lengths of the objective and eyepiece.

4.4 Telescope

A telescope is very similar to a compound microscope except the object is at infinity and thus the objective lens forms an image at the objective focus. Note that, more often than not, we set our telescope to the orientation where our eyes are most relaxed and final virtual

object is still formed at infinity, just bigger. Nevertheless, we can bring the moon to our near point too.

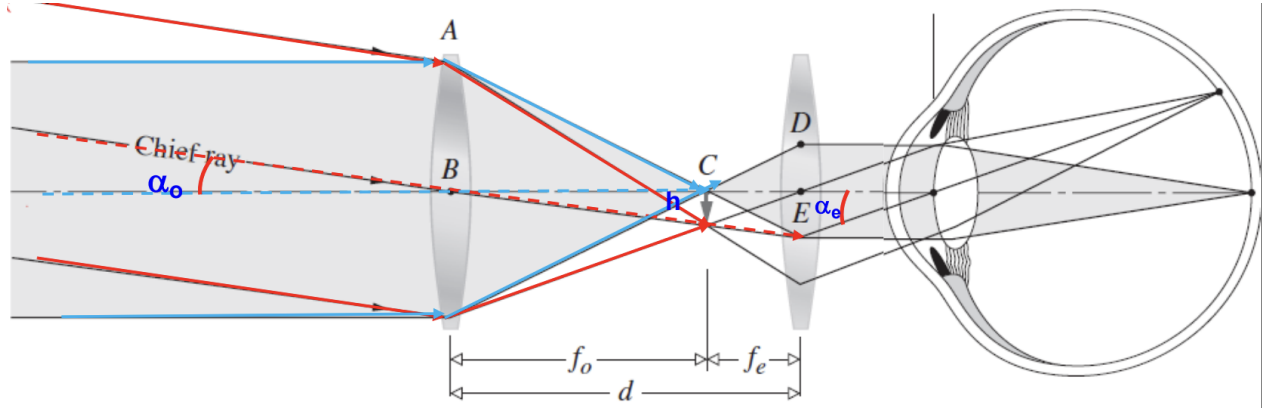


Figure 7: Telescope with objective and eyepiece lenses

In the above diagram, assume that the blue rays are coming from the bottom part of our astronomical object and red rays are coming from the top. In the absence of any telescope, α_0 is the angle subtended in our eyes. But in the presence of a telescope, there is an angular magnification given by

$$M_{\infty} = \frac{\alpha_e}{\alpha_0}$$

Also, in this case, I can write $\alpha_0 \approx h/f_0$ and $\alpha_e \approx h/f_e$. Combining them both, I arrive at the final result

$$M_{\infty} = \frac{f_0}{f_e} \quad (23)$$

Here too, we can define the magnification for the final virtual object at near point. That is,

$$M_{\text{near point}} = \frac{f_0}{f_e} \left(1 + \frac{f_e}{D} \right)$$

4.5 Prism

A prism is mostly used for dispersion of light. The main principle behind this is the fact that different wavelengths of light have different refractive indices in the same medium and thus they refract at different angles at the first interface. The reason why a glass prism is able to disperse light while a glass slab cannot even though both possess the same kind of refractive index is because of how the incident face and emergent face of a prism are not parallel and this enhances the dispersive ability. In case of glass, the different wavelengths emerge with the same angle of refraction. A simple proof of this would be the fact two prisms placed side-by-side upside down gives glass slab. The first prism's effect is canceled by the second prism.

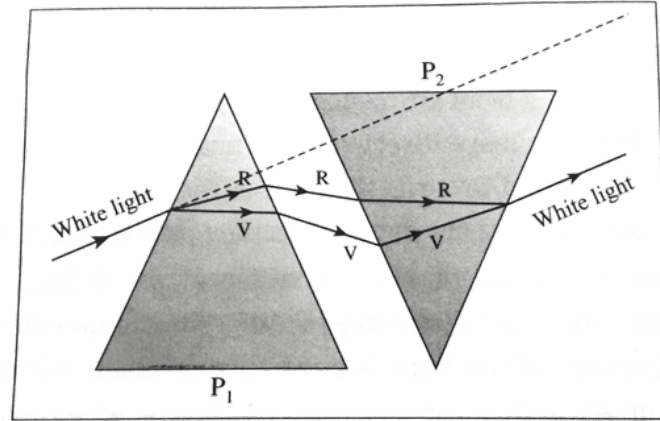
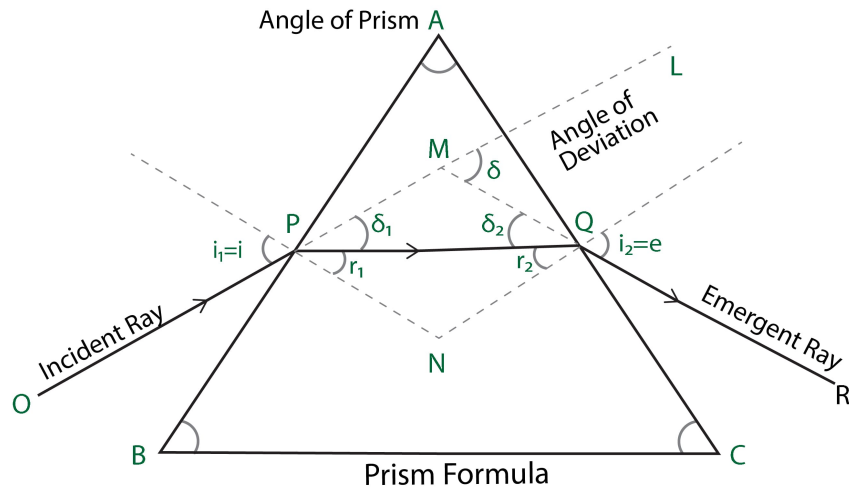


Figure 8: A glass slab is a combination of two prisms

In a way, we can say that a prism is able to disperse light because it is able to deviate the light from its original angle and different wavelengths have different angles of deviation. We shall now study the mathematics for the deviation of light and how it depends on the refractive index. For that, consider the prism given below



The angle formed by the incident face and the emergent face is called the angle of the prism – $\angle A$. i is the angle of incidence, e is the angle of emergence and r_1 and r_2 are the angles made by the light-ray inside the prism. In quadrilateral $APNQ$, we can see that

$$\begin{aligned}\angle A + 90^\circ + \angle N + 90^\circ &= 360^\circ \\ \implies \angle A + \angle N &= 180^\circ\end{aligned}$$

In triangle, $\triangle PNQ$, we see that

$$\begin{aligned} r_1 + r_2 + \angle N &= 180^\circ \\ \implies r_1 + r_2 + (180^\circ - \angle A) &= 180^\circ \\ \implies r_1 + r_2 &= \angle A \end{aligned} \quad (24)$$

From this, we see that

$$(i - \delta_1) + (e - \delta_2) = \angle A$$

and since the total deviation $\delta = \delta_1 + \delta_2$ (by exterior angle property)

$$\delta = i + e - \angle A \quad (25)$$

From Snell's Law, I know that $n = \frac{\sin(i)}{\sin(r_1)} = \frac{\sin(e)}{\sin(r_2)}$. If we use small angle approximation here too, we get $i \approx nr_1$ and $e \approx nr_2$. Substituting this in (25), I get

$$\begin{aligned} \delta &= nr_1 + nr_2 - \angle A \\ &= n(r_1 + r_2) - \angle A \\ &= n\angle A - \angle A \end{aligned}$$

Thus for small angles of incidence, we get

$$\boxed{\delta \approx (n - 1)A} \quad (26)$$

Clearly, unlike a glass slab, no matter the angle of incidence, there is always a deviation angle and this deviation angle depends on the angle of incidence, refractive index and angle of the the prism. Therefore, there is also an angle for which, this deviation angle is minimum. An important thing to note here is that **there can only be one incident angle for which minimum deviation happens and it must be the same if the light ray is sent from the other side too**. Thus, when minimum deviation happens, the incident ray and emergent ray are symmetric, and also the ray inside the matrix will be perfectly horizontal (all to respect left-right symmetry). Thus, when minimum deviation happens, $i = e$, $\delta_1 = \delta_2$ and $r_1 = r_2$. Then,

$$\delta_{\min} = 2i - \angle A \quad \text{and} \quad 2r = \angle A$$

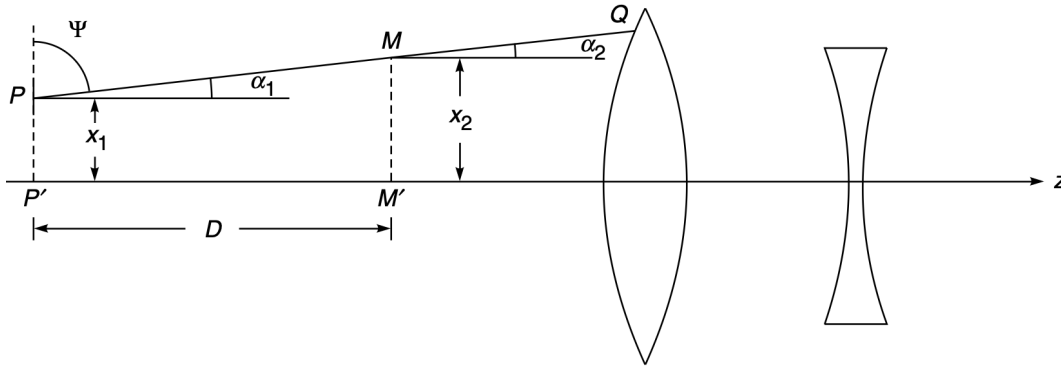
Since, the refractive index of the glass prism is $n = \frac{\sin(i)}{\sin(r)}$, I can write it as

$$\boxed{n = \frac{\sin\left(\frac{A + \delta_{\min}}{2}\right)}{\sin\left(\frac{A}{2}\right)}} \quad (27)$$

5 Matrix Method

When you do matrix method in Optics, you are essentially taking an initial set of parameters of a ray of light, applying a linear operator (which is our matrix of interest) on it and getting the final set of parameters. Matrix method is particularly useful when you have a number of lenses. You don't have to manually look at where the image formed by each lens is and make that the virtual object for the next lens. Instead, you can just stack one matrix over another until you take care of every lens in the problem.

Let's think of a ray of light emerging from a point, directed towards an optical system (say lens) at some angle.



Two good parameters to describe our ray of light at any point is its height above the principal axis – x , and the angle it makes with the principal axis – α . Let's first see how these parameters evolve with time (or more specifically D) for transmission of light in a homogeneous medium. For that let's study how the light's parameters evolved from point- P to point- M .

Firstly, light travels along in a straight line in a medium. Therefore α doesn't vary during transmission ie. $\alpha_2 = \alpha_1$. On the other hand, the height of the ray above the principal axis does vary because of the initial angle α_1 and by constructing a triangle, we can see that

$$x_2 = x_1 + D \tan(\alpha_1)$$

We are still only considering paraxial rays and therefore, we can approximate $\tan(\alpha_1) \approx \alpha_1$.

$$\implies x_2 = x_1 + \alpha_1 D$$

Instead of dealing with angles, I want to introduce a quantity – 'Optical Directional Cosine' (λ) which is the product of refractive index of the medium and sine of the angle α . This will come useful because we are eventually going to be dealing with refraction. Using the paraxial approximation again, I can write $\lambda_1 \approx n_1 \alpha_1$. The earlier equations would then become

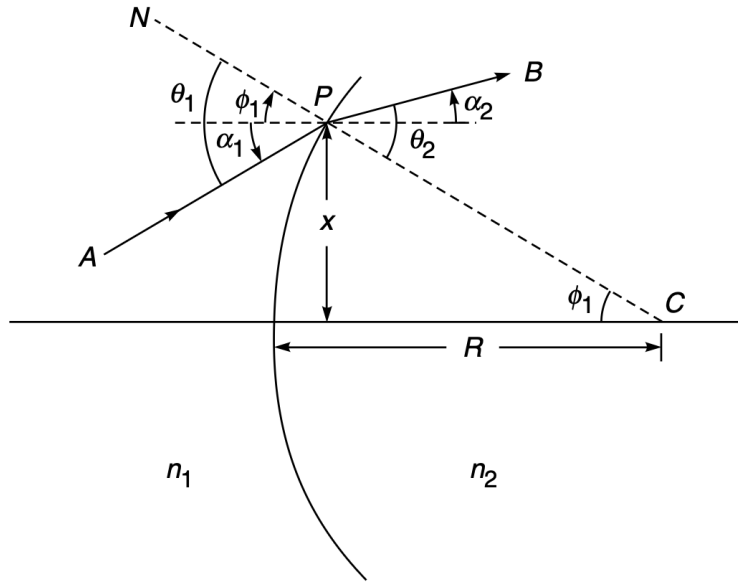
$$x_2 = x_1 + \frac{\lambda_1 D}{n_1} \quad \text{and} \quad \lambda_2 = \lambda_1$$

I can write the above result in the form of a linear transformation using a matrix

$$\begin{bmatrix} \lambda_2 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \frac{D}{n_1} & 1 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ x_1 \end{bmatrix} \quad (28)$$

The 2×2 matrix in the above equation is called the ‘**Transmission-Matrix**’ for a ray.

Now, let's work out what will happen when this ray of light encounters a medium interface.



In this case, the quantity that remains same just before just after refraction is the height of the ray above the principal axis i.e. $x_2 = x_1 = x$. By Snell's law,

$$\begin{aligned} n_1 \sin(\theta_1) &= n_2 \sin(\theta_2) \\ n_1 \sin(\alpha_1 + \phi_1) &= n_2 \sin(\alpha_2 + \phi_1) \end{aligned}$$

Applying paraxial approximations — $\sin(\theta) \approx \theta$ and $\phi_1 = \frac{x}{R}$, I get

$$\begin{aligned} n_1(\alpha_1 + \phi_1) &= n_2(\alpha_2 + \phi_1) \\ \implies n_2\alpha_2 &= n_1\alpha_1 - \frac{(n_2 - n_1)}{R}x \end{aligned}$$

The quantity $(n_2 - n_1)/R$ is called the ‘Power of a Refracting Surface’(P). Hence, I again get two equations

$$n_2\alpha_2 = n_1\alpha_1 - Px \quad \text{and} \quad x_2 = x_1$$

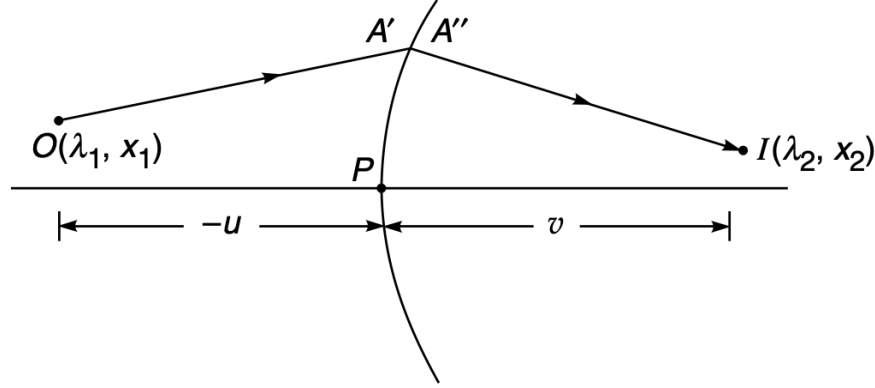
And once again, I can write this in matrix notation

$$\begin{bmatrix} \lambda_2 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 & -P \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ x_1 \end{bmatrix} \quad (29)$$

Like before, the 2×2 matrix in this equation is called the ‘**Refraction-Matrix**’ for a ray.

5.1 For a spherical Interface

The scenario of a spherical interface has 3 parts – (1) Light traveling towards the interface, (2) Light refracting and (3) Light traveling after the refraction.



Let the initial parameters at O be (λ_1, x_1) and the final parameter at I be (λ_2, x_2) . Just before the refraction the parameters are (λ', x') and just after the refraction the parameters are (λ'', x'') . Then,

1. Transmission from $O \rightarrow A'$ is given by

$$\begin{bmatrix} \lambda' \\ x' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \frac{-u}{n_1} & 1 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ x_1 \end{bmatrix}$$

2. Refraction from $A' \rightarrow A''$ is given by

$$\begin{bmatrix} \lambda'' \\ x'' \end{bmatrix} = \begin{bmatrix} 1 & -P \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \lambda' \\ x' \end{bmatrix}$$

3. Transmission from $A'' \rightarrow I$ is given by

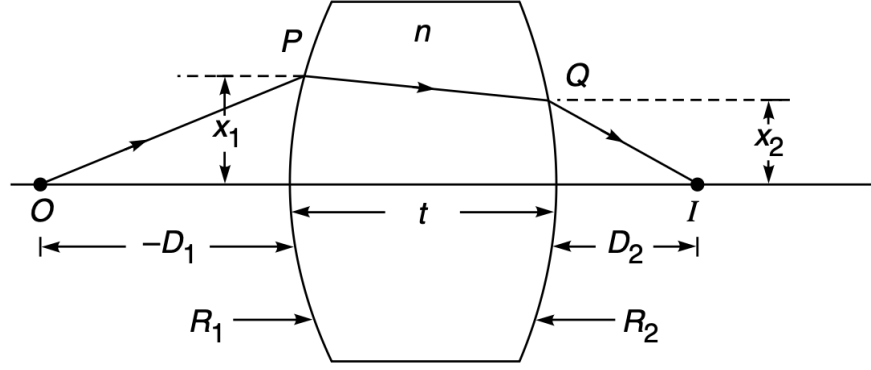
$$\begin{bmatrix} \lambda_2 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \frac{v}{n_2} & 1 \end{bmatrix} \begin{bmatrix} \lambda'' \\ x'' \end{bmatrix}$$

Combining all 3, I get

$$\begin{bmatrix} \lambda_2 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \frac{v}{n_2} & 1 \end{bmatrix} \begin{bmatrix} 1 & -P \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ \frac{-u}{n_1} & 1 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ x_1 \end{bmatrix} \quad (30)$$

5.2 For Lens

For lenses, we take the example of a point object on the principal axis. It forms a point image also on the principal axis.



$$\begin{bmatrix} \lambda_2 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 & -P_2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ \frac{t}{n} & 1 \end{bmatrix} \begin{bmatrix} 1 & -P_1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ x_1 \end{bmatrix} \quad (31)$$

This might look incomplete since I haven't taken into consideration the transmission part from object to lens and from lens to image. The reason for that is that I don't need to calculate those using matrix. Since both initial and final x 's are 0, simple trigonometric extensions will give me what I need.

In the case of a thin lens, we just ignore transmission matrix inside the lens.